

VL5T24EC211CO-2 AIM-1

3) Consider the following CMOS inverter circuit which consists of two enhancement-type NMOS transistor, with parameter

$$V_{T0} = 0.8V$$

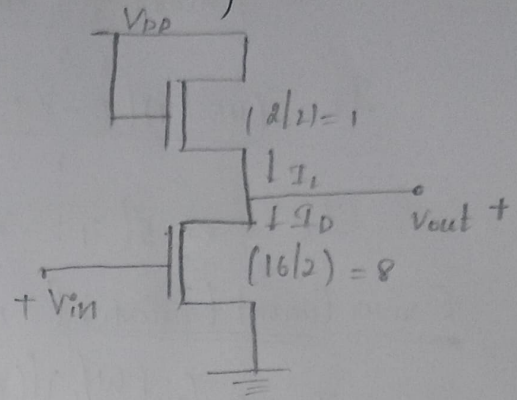
$$\mu_n C_{ox} = 45.0 \mu A/V^2$$

$$\gamma = 0.38 V^{1/2}$$

$$|\phi_F| = 0.6V$$

$$\lambda = 0$$

$$V_{DD} = 5.0V$$



1) Calculate V_{OH} and V_{OL} values. Note the substrate bias effect of the load device must be taken into consideration!

- Here $V_{in} = 0$ so PDN is completely to zero
- load transistor bulk is at ground
- so let $V_{SB} = V_{OH} \rightarrow (1)$

We know that threshold Voltage

$$V_T = V_{T0} + \gamma (\sqrt{|\phi_F| + V_{OH}} - \sqrt{|\phi_F|}) \rightarrow (2)$$

At steady state mean cutoff ($V_{GS} = V_T$) $\therefore V_{GS} = V_{DD} - V_{OH}$

$$5 - V_{OH} = 0.8 + 0.38 (\sqrt{0.6 + V_{OH}} - \sqrt{0.6})$$

$$5 - V_{OH} = 1.18 (\sqrt{0.6 + V_{OH}} - \sqrt{0.6})$$

$$3.82 - V_{OH} = (\sqrt{0.6 + V_{OH}} - \sqrt{0.6})$$

$$\boxed{V_{OH} = 4.1V}$$

Output Low Voltage V_{OL} :

Where $V_{in} = V_{OH} \rightarrow (1)$

Driver, load transistor = ON (saturation, triode)

Load Current saturation ($I_L = I_D$)

$$\therefore V_{G1S} = V_{DD} - V_t$$

$$I_D = \frac{\mu_n \epsilon_0 x \left(\frac{W}{L} \right) (V_{G1S} - V_t)^2}{2}$$
$$= \frac{\mu_n \epsilon_0 x \left(\frac{W}{L} \right) (V_{DD} - V_{OL} - V_t)^2}{2}$$

$$I_L = \frac{(45)(1)(5 - V_{OL} - 0.8)^2}{2} = \frac{(45)(1)(4.2 - V_{OL})^2}{2}$$

$$I_L = 22.5 (4.2 - V_{OL})^2 \rightarrow (1)$$

Driver Current (triode) ($I_L = I_D$)

$$I_D = \mu_n \epsilon_0 x \left(\frac{W}{L} \right) \left[(V_{G1S} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$= 45 \times 8 \left[(4.1 - 0.8) V_{OL} - \frac{V_{OL}^2}{2} \right]$$

$$= 360 [3.3 V_{OL} - 0.5 V_{OL}^2] \rightarrow (2)$$

Equate (1) = (2)

$$22.5 (4.2 - V_{OL})^2 = 360 [3.3 V_{OL} - 0.5 V_{OL}^2]$$

$$\boxed{V_{OL} = 0.2V}$$

b) Interpret the result in term of noise margin and static DC power dissipation?

→ Noise Margin

1) $V_{DD} = 5V$ Actual High = 4.1V Noise margin is moderate

2) $V_{DD} = 5V$ Actual low = 0.2 Noise margin is good

→ static power

$$\text{Input} = 0$$

$$P_{DC} = 0$$

$$\text{Input} = 1$$

$$P_{DC} = V_{DD} \times I_{DD}$$

c) calculate steady current which drawn from DC power supply when input logic 1?

We know that $V_{in} = V_{OH}$

$$\text{Where } I_{DD} = I_L = I_D \rightarrow (1)$$

$$\therefore I_L = 0.36 \mu A \text{ to } 0.8 mA \quad \boxed{\text{so } I_D = 0.5 mA}$$